MATTHEW D. HALL

mattdanhall.com
mattdanhall@duck.com

Postgraduate researcher dedicated to finding novel ways to advance the future of automation. Expertise in designing, developing, evaluating and presenting control algorithms using custom built simulation software. Proven track record of effectively communicating complex and varied concepts to audiences of differing backgrounds. Seeking a new challenge in an ambitious, mission-driven company which is aiming to foster novelty and innovation at their core.

COMPETENCIES & TECH SKILLS

Modular & Swarm Robotics • Algorithm Design • Software Development • Data Analysis • Project Management • Technical Writing/Presentation • LaTeX • Python • NumPy • Pandas • Jupyter • C/C++ • Git • CI/CD • TDD • Linux • HPC • Shell Scripting

EXPERIENCE

Postgraduate Researcher, PhD - *Automatic Control and Systems Engineering* 2017 - 2022

Developed and verified novel control algorithms for the control of modular, multi-robot systems. Created custom simulation and evaluation software using Python to verify work. Responsible for the entire research cycle, from idea generation to scientific reporting. Collaborated with academics throughout the process, including as a mentor to other research students.

Peer Reviewer - *Natural Robotics Lab* 2017 - 2021

Reviewed submitted works for a number of highly regarded international conferences, such as: AAMAS, ANTS, DARS, IROS, MRS, RSS and TAROS. Necessary to keep up to date with state of the art work. Utilised my ability to disseminate information and assess the quality of written works.

Graduate Teaching Assistant - *The University of Sheffield* 2017 - 2020

Assisted in the planning and running of laboratory based teaching and seminar sessions for Master's and Bachelor's students. Required an in-depth knowledge of the material and the ability to clearly communicate to answer the students' questions. Organisational skills also required to effectively plan sessions for multiple groups for the semester.

Tourism Advisor - *Use-it Brussels* 2016 - 2017

Part-time work within a sustainable tourism project while residing in Belgium, offering tourist information and running tours of the lesser-known parts of Brussels. A great opportunity to thoroughly learn about the city and its culture, as well as improving my interpersonal skills.

Team Captain & Coach - *Lancaster Ligers Korfball Team* 2015 - 2016

Earned Level 2 coaching and referee qualification in Aug 2015. Assisted with training and squad selection. Also responsible for fixture and travel arrangements. Invaluable in furthering team-leadership skills.

Student Ambassador - *Lancaster University* 2014 - 2016

Part-time employment assisting with university events and the regular running of tours around the university campus. These responsibilities required excellent communication and organisational skills, and a personable yet professional demeanour.

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EDUCATION

PhD - ***Automatic Control and Systems Engineering*** 2017 - 2022
The University of Sheffield - **In Review**

THESIS: “Active Subtraction: A Viable Method of Self-Reconfiguration for Modular Robotic Systems”

OVERVIEW: The creation, verification and demonstration of a number of novel control algorithms to guarantee autonomous reconfiguration of multi-robot modular systems, enabling modules to remove themselves from a given starting configuration to form any desired shape.

MEng (Master of Engr.) - ***Electrical and Electronics Engineering*** 2012 - 2016
Lancaster University - **Awarded 1st Class Hons**

A-Levels - ***Maths (A*) Further Maths (A) Physics (B)*** 2010 - 2012
Leeds, Abbey Grange Academy Sixth Form - **Governor’s Award for Outstanding Results**

PUBLICATIONS

M. D. Hall, A. Özdemir, R. Groß, **“Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots”**, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020. **(Presented virtually: https://youtu.be/RgGi_mc4tOQ)**

A. Özdemir, M. Gauci, A. Kolling, **M. D. Hall**, R. Groß, **“Spatial Coverage Without Computation”**, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pages 9674–9680. **(Presented in person)**

AWARDS

EPSRC Doctoral Training Partnership National Productivity Scholarship for course and maintenance fees · University Postgraduate Research Student Publication Scholarship for 3 months funding towards journal publication · Best Oral Presentation at Departmental Symposium · Best Poster Presentation at Faculty Researcher Symposium

FURTHER INTERESTS

PROGRAMMING: Proficient in **Python**, **C/C++**, **Git**, **Linux**, **LaTeX**, **Shell Scripting**.
Working knowledge of **MATLAB**, **Java**, **HTML**, **CSS** and **VHDL**.
Designed and created my personal website using the **Hugo** framework.
Released a small number of customisation apps for KLWP on **Android**.
Selection of other projects available on my **GitLab**.

MUSIC: Bassist and guitarist. Experience in Ableton music production.
Regular guest **radio DJ** for “Clear A Space” on Gaga Radio.

TRAVEL: Enjoy learning about cultures and being immersed in new cities.
Lived and worked in Brussels following Master’s degree completion.

FILM: Keen interest in film. **Written**, **shot** and **edited** amateur films with family and friends.

OTHER: Food and cooking, games (board or video), craft beer.

REFERENCES AVAILABLE UPON REQUEST